

Multiple Choice (10 marks)

1. Which of the following best describes what discriminative approaches try to model? (w are the parameters in the model)
 - (a) $p(y|x, w)$
 - (b) $p(y, x)$
 - (c) $p(w|x, w)$
 - (d) None of the above
2. Which one of the following is equal to $P(A, B, C)$ given Boolean random variables A, B and C, and no independence or conditional independence assumptions between any of them?
 - (a) $P(A | B) * P(B | C) * P(C | A)$
 - (b) $P(C | A, B) * P(A) * P(B)$
 - (c) $P(A, B | C) * P(C)$
 - (d) $P(A | B, C) * P(B | A, C) * P(C | A, B)$
3. Averaging the output of multiple decision trees helps __.
 - (a) Increase bias
 - (b) Decrease bias
 - (c) Increase variance
 - (d) Decrease variance
4. After applying a regularization penalty in linear regression, you find that some of the coefficients of w are zeroed out. Which of the following penalties might have been used?
 - (a) L_0 norm
 - (b) L_1 norm
 - (c) L_2 norm
 - (d) either (a) or (b)
5. If N is the number of instances in the training dataset, nearest neighbors has a classification run time of
 - (a) $O(1)$
 - (b) $O(N)$
 - (c) $O(\log N)$
 - (d) $O(N^2)$

True / False (10 marks)

Answer True or False

6. True or False: The model obtained through best-subset selection will always match the model derived from forward stepwise selection.
7. True or False: Lasso is more effective than Ridge for feature selection due to its ability to shrink some coefficients to zero.
8. True or False: The convergence of a neural network is influenced by the learning rate, affecting how quickly it learns from data.
9. True or False: Clustering is the best approach for predicting the amount of rainfall based on various environmental cues.
10. True or False: The BLEU metric evaluates translation quality using precision, while the ROGUE metric focuses on recall in summarization tasks.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the demlo number for the given number.
12. Write a python function to find the sum of even factors of a number.

End of Paper